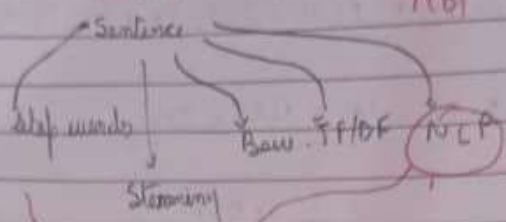


22-09-2025

Tutorial - 49

$$P(A/B) = P(B/A) + \frac{P(A)}{P(B)}$$



↳ Preprocessing steps for NLP

↳ Natural Language Processing

sentences $[x_1, x_2, x_3, x_4, \dots]$

$P(y = \text{yes} | \text{Sentences})$

$\Rightarrow P(y = \text{yes} | (x_1, \dots, x_n))$

$$P(y) \times \prod_{i=1}^n P(x_i | y = \text{yes}) \times P(x_2 | y = \text{yes}) \times \dots \times P(x_n | y = \text{yes})$$

f_1	f_2	f_3	f_4	O/P
The	food	Delicious	Bad	1
1	1	1	0	1
1	1	0	1	0
0	1	0	1	0
0	1	1	0	1
0	0	0	1	0

will be removed by NLP

Sent 1 $\rightarrow$ The food is delicious

Sent 2 $\rightarrow$ The food is Bad

Sent 3 $\rightarrow$ Food is Bad

$\rightarrow$ stop words

PTO

$$P(y = \text{Yes} | \text{Sentences})$$

$$\Rightarrow P(y = \text{Yes}) \times P(x_1 / y = \text{Yes}) \times P(x_2 / y = \text{Yes}) \times P(\dots)$$

$$\Rightarrow \frac{2}{5} \times \frac{1}{2} \times \frac{2}{4} \times \frac{2}{2}$$

$$\Rightarrow \frac{1}{10} \Rightarrow 0.1$$

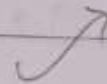
$$P(y = \text{No} | \text{Sent})$$

$$\Rightarrow 0.03$$

$$\frac{3}{5} + \frac{1}{2}$$

Normalizing into 1

$$\Rightarrow \frac{0.01}{0.01 + 0.03} \Rightarrow \frac{1}{4} = 0.25\%$$



Yes

$$\text{No} \Rightarrow 0.75$$